

Weekly Progress Report

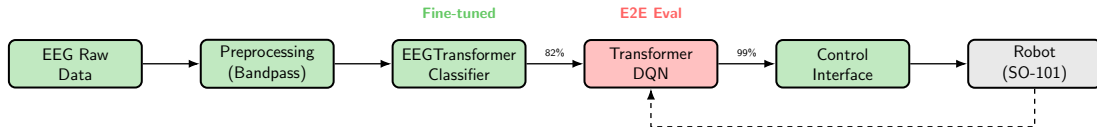
Week 26: Offline E2E Evaluation & Final Report Draft Completion

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BCI Control System Project

March 17, 2026

System Overview: Current Focus



Legend:

- Completed
- In Progress

- This Week's Focus
- Pending (Hardware)

This Week's Objectives

Goals

- 1 CTNet + DQN offline end-to-end evaluation (simulated)
- 2 EEG-aware RL training comparison (with/without EEG input)
- 3 ICA preprocessing comparison experiment
- 4 Update final report with all experimental results (v4 $\rightarrow$ v6)

Key Question to Address

Can the RL agent compensate for imperfect EEG classification (82%) to achieve reliable robotic control?

Background: PhysioNet CTNet Training

Dataset: PhysioNet EEG Motor Movement/Imagery

- 109 subjects, 64 channels, 160 Hz
- 4-class MI: Left hand, Right hand, Both feet, Both fists
- ~90 trials per subject

Model: EEGTransformer (Hybrid CNN + Transformer)

- EEGNet-style conv front-end for spatial filtering
- Multi-layer Transformer encoder for temporal modeling
- Residual fusion + classification head

Results:

Configuration	Accuracy
Cross-subject (109) + Fine-tuning	56.54% 88.78%
Improvement	+32.24pp

Per-Subject Fine-tuned Range:

- Best: 97.78% (S059)
- Worst: 75.56% (S088)
- Reflects individual MI ability

Significance

Transfer learning enables fast adaptation with minimal calibration data

Background: Channel Reduction Study

Motivation:

- OpenBCI Cyton only has 8 channels
- Need to reduce from 64 $\rightarrow$ 8 for deployment
- Which channels are most important?

Methodology:

- Systematic ablation: 64 $\rightarrow$ 32 $\rightarrow$ 16 $\rightarrow$ 8
- Channel importance ranking via accuracy drop
- Focus on motor cortex region (C3, C4, Cz)

Key Finding:

C3 is Critical

Removing C3 causes largest accuracy drop (-8.3%)
Consistent with left motor cortex role in MI

Results by Channel Count:

Channels	Cross-Sub	Fine-tuned
64 (full)	56.54%	88.78%
32	54.21%	85.43%
16	51.87%	80.12%
8	48.92%	72.54%

Selected 8 Channels:

C3, C4, Cz, FC3, FC4, CP3, CP4,
Pz

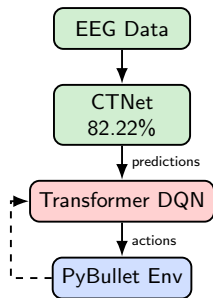
Conclusion

8-channel config retains 72.54% accuracy
Viable for OpenBCI deployment

Experiment 1: Offline E2E Evaluation (Simulated)

Methodology:

- 1 Load pre-trained CTNet (PhysioNet 109 subjects)
- 2 Classify test subjects' EEG $\rightarrow$ 82.22% accuracy
- 3 Train Transformer DQN with/without EEG predictions
- 4 Evaluate control in **PyBullet simulation** (not physical robot)



Two Conditions:

- **Arm state only:** 6D joint positions
- **Arm + EEG:** 6D + 4 class probs + confidence

Experiment 1: Results – RL Compensates in Simulation

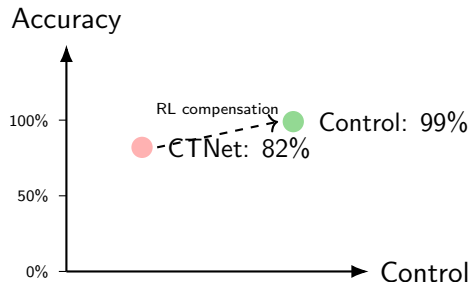
Performance Comparison:

Metric	Arm Only	Arm + EEG
Reach Rate	99.0%	98.7%
Mean Reward	91.2	90.8
Mean Steps	18.7	19.1
Train Time	1130s	668s
Speedup	–	41%

Key Insight:

- Both achieve $\sim 99\%$ reach rate
- EEG input reduces training time by **41%**
- RL compensates for $82\% \rightarrow 99\%$ control

Classification $\rightarrow$ Control



*RL learns to handle
classification uncertainty*

Experiment 2: ICA Preprocessing Comparison

Methodology:

- Dataset: BCI Competition IV-2a (MAT files)
- Compare: with ICA vs. without ICA
- ICA: FastICA, remove 2 highest-variance components
- 3 subjects tested (S1, S3, S7)

Results:

Subject	No ICA	With ICA	Diff
S1	77.78%	75.00%	-2.78%
S3	86.11%	87.50%	+1.39%
S7	69.44%	75.00%	+5.56%
Avg	77.78%	79.17%	+1.51%

Key Findings:

Marginal & Inconsistent

- Average gain: only +1.51%
- S1 actually **degraded**
- 8–30Hz bandpass already removes most ocular artifacts

Decision

Exclude ICA from standard pipeline

- Adds complexity
- Inconsistent benefit
- Risk of removing signal

Transfer Learning: Fine-tuning Mechanism

Two-Stage Training:

Stage 1: Cross-Subject Pre-training

- Train on 109 subjects (PhysioNet)
- Learn population-level MI patterns
- Result: 56.54% cross-subject accuracy

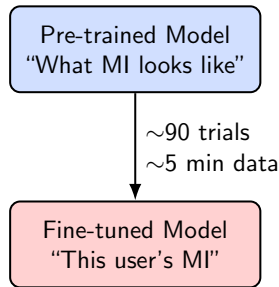
Stage 2: Subject-Specific Fine-tuning

- Adapt to individual's neural signature
- Lower learning rate (1×10^{-4})
- Only ~ 90 trials needed
- Result: **88.78%** accuracy (+32pp)

Why Fine-tuning Works

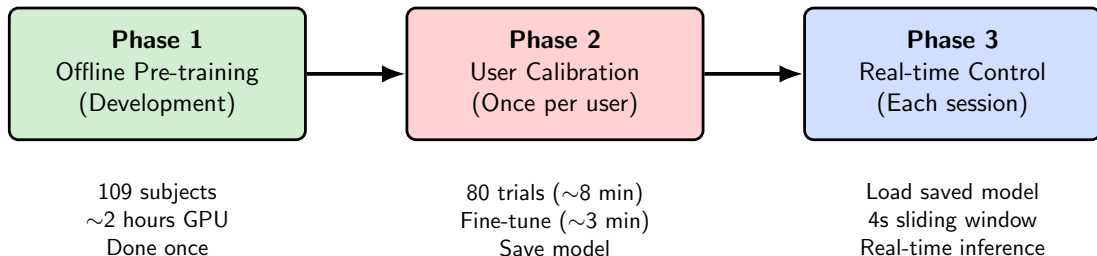
General:
Mu/Beta patterns
Spatial distribution
Temporal dynamics

Individual:
C3/C4 location
Cognitive strategy
Impedance variations



*Model already knows MI structure;
only needs to adapt to individual*

Deployment Workflow: Three Phases



Total Setup Time per User:

- Data collection: ~8 minutes
- Fine-tuning: ~3 minutes (GPU)
- **Total: ~11 minutes**

Comparable to Commercial BCIs:

- Emotiv: 10–15 min calibration
- NeuroSky: 5–10 min setup
- Our system: 11 min (one-time)

Final Report: v4 → v6 Updates

v4 → v5 (Fine-tuning Details):

- Training Strategy: two-stage protocol
- Real-Time Pipeline: three-phase deployment
- Discussion: why fine-tuning works
- Limitations: 10-subject validation note

v5 → v6 (Hardware Status):

- New section: “Hardware Validation Status”
- Clarify what's completed vs. pending
- Updated conclusion with next steps

Hardware Validation Status:

Completed (Software)

- EEGTransformer on 3 datasets
- DQN in PyBullet (99% reach)
- BrainFlow API integration
- SO-101 serial protocol

Pending (Hardware)

- Live human subject testing
- OpenBCI + SO-101 closed loop
- Scheduled with supervisor

Summary: All Experimental Results

Experiment	Key Finding	Status
Cross-subject CTNet	56.54% (109 subjects)	✓
Fine-tuned CTNet	88.78% (+32pp)	✓
Channel Reduction	8-ch: 72.54% (viable)	✓
Simulated E2E Control	82% class → 99% control	✓
EEG-aware Training	41% faster training	✓
ICA Comparison	+1.51% (excluded)	✓
Physical Robot Test	Pending hardware	○

All software experiments completed.
Hardware validation is the only remaining task.

Next Steps: Hardware Validation

Priority Tasks (Pending Hardware Access)

- 1 Validate 8-channel OpenBCI configuration with live subject
- 2 Measure real-time classification latency
- 3 End-to-end closed-loop test: EEG $\rightarrow$ CTNet $\rightarrow$ DQN $\rightarrow$ SO-101
- 4 Qualitative usability assessment

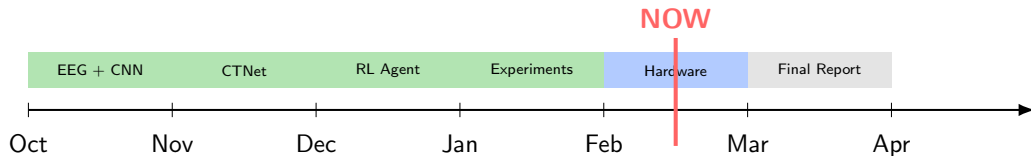
Hardware Requirements:

- OpenBCI Cyton (8-channel)
- SO-101 robotic arm
- Participant recruitment
- Safety protocol approval

Questions for Supervisor:

- Hardware session?
- Participant recruitment process?
- Safety/ethics approval status?

Project Timeline



Current Status: On track – all software work completed

This Week's Achievement: Offline E2E evaluation shows 82% → 99% control (simulated)

Key Achievements This Week

- ① **Offline E2E Evaluation:** RL compensates 82% classification → 99% control (simulated)
- ② **EEG-aware Training:** 41% faster convergence with EEG input
- ③ **ICA Decision:** Excluded from pipeline (marginal +1.51%, inconsistent)
- ④ **Fine-tuning Analysis:** ~11 min calibration enables 88.78% accuracy
- ⑤ **Report v6:** Complete with hardware status clarification

*All software experiments completed.
Ready for hardware validation when equipment available.*